

# Nishant Choudhary

B.Tech Electronics & Computer Engineering | VIT Chennai | 2027

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## EXPERIENCE

### Mimic Productions — AI Programming Intern

July 2026 – Present

Python · Pillow · NumPy · rembg/U<sup>2</sup>-Net · fal.ai · OpenAI API

Remote

- Built a Python pipeline rendering brand marketing carousels entirely in code — 12 layouts × 4 colour treatments (**48 looks**) at **~1 s/image** and zero marginal cost, lifting a hard ~7-image-per-session ceiling.
- Replaced a failing colour-key background removal with **U<sup>2</sup>-Net segmentation**, adding premultiplied-alpha compositing and cache versioning to eliminate a class of silent stale-asset defects.
- Benchmarked image-generation providers (OpenAI gpt-image-1 vs FLUX via fal.ai) and found **context-gathering calls, not generation, dominated API cost** — redirecting the team from a provider switch to caching, with a substantially larger projected saving.

## SELECTED PROJECTS

### Government Budget Analytics Platform

🐙 GitHub

Python · MySQL 8 · Streamlit · pandas

- Cut default page load of a **922K-row** (921.5K synthetic) MySQL analytics dashboard from **3m37s to 30s (7.2×)** by profiling what the application actually issued and finding **99.5%** of the time in two queries that no index could reach.
- Proved the cause from query plans rather than wall-clock: MySQL will not push a **disjunctive** predicate into a GROUP BY view, so IN/OR materialise **201,714** rows where a UNION ALL of equalities materialises **80**.
- Measured a covering index the docs credit with 80–85% at **0.8%** — inside run-to-run noise — and published the negative result; the real win was buffer-pool sizing at **32.7%**, mediated by an adaptive hash index MySQL's defaults make *net-harmful*.
- Reproduced and fixed two concurrency defects — a lost-update race under REPEATABLE READ, and a shared connection that segfaulted under concurrent readers — plus an ETL grain bug discarding **66% of actuals**.

### ClickTutor AI

🐙 GitHub · 📝 Writeup

Python · PyQt6 · Gemini API · Tesseract

- Built a Python/PyQt6 screen-overlay tutor: **228 tests**, CI (ruff + mypy + pytest) green on every push, every benchmark number reproducible from committed JSON with its recorded environment.
- Fixed a coordinate-mapping defect rendering highlights **entirely outside the overlay widget** under display scaling — OCR reports physical pixels, Qt logical ones — by extracting the transform into a **pure, Qt-free module** unit-tested across DPI scaling and letterboxing.

### Fake News Detection REST API

🌐 Live Demo · 🐙 GitHub

Python · FastAPI · BERT · ONNX Runtime · Docker

- Deployed a FastAPI + Docker service around a fine-tuned BERT 6-class LIAR classifier at **8.5 ms mean / 12.8 ms P95** and **396 req/s** batched, with every published figure regenerated by a committed script.
- Benchmarked **ONNX Runtime** export and int8 quantisation on a **separate model-level harness** (torch fp32 **236 ms P95** → int8 **86 ms**): kept fp32, **rejected int8** despite 2.7× lower P95 — McNemar **p=0.38** left the drop insignificant, but only **57%** of predictions matched, so equivalence was unproven.

## TECHNICAL SKILLS

|                  |                                                                                                                                                    |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages & Data | Python, C++, SQL/MySQL (Schema Design, Query Optimization, Indexing, ETL)                                                                          |
| Engineering      | REST APIs, FastAPI, Docker, Git/GitHub, GitHub Actions CI, Linux, Software Architecture, Profiling & Benchmarking, Concurrency, pytest, ruff, mypy |
| AI / ML          | Deep Learning, Computer Vision, NLP, Transformers, Transfer Learning, Diffusion Models, Image Segmentation, Quantisation                           |
| Frameworks       | PyTorch, TensorFlow, Scikit-learn, OpenCV, Hugging Face, ONNX Runtime, Streamlit, Pandas, NumPy                                                    |

## EDUCATION

B.Tech — Electronics & Computer Engineering, VIT Chennai CGPA: 8.32 / 10 Aug 2023 – May 2027

## ACHIEVEMENTS & CERTIFICATIONS

- 150+** DSA problems solved on LeetCode · **Advanced Generative AI**, IBM Career Education Program (Jul 2026) · **Oracle OCI AI Foundations Associate** (2025) · **Machine Learning Specialization**, Andrew Ng (Coursera).